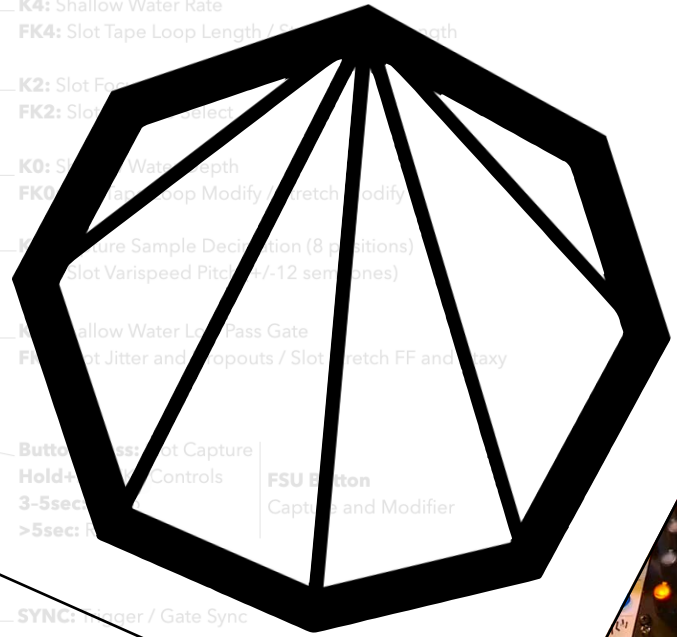
MOTUS DIVERSI VERSIO

3x MEMORY LOOPER FIRMWARE FOR THE NE VERSIO PLATFORM

JASON VON HOCKMAN





GTNP

SODA

Manchester Metropolitan University

OVERVIEW

PART I: INTRODUCTION

Background
Memory / Granular Synthesis
Timbral Modification

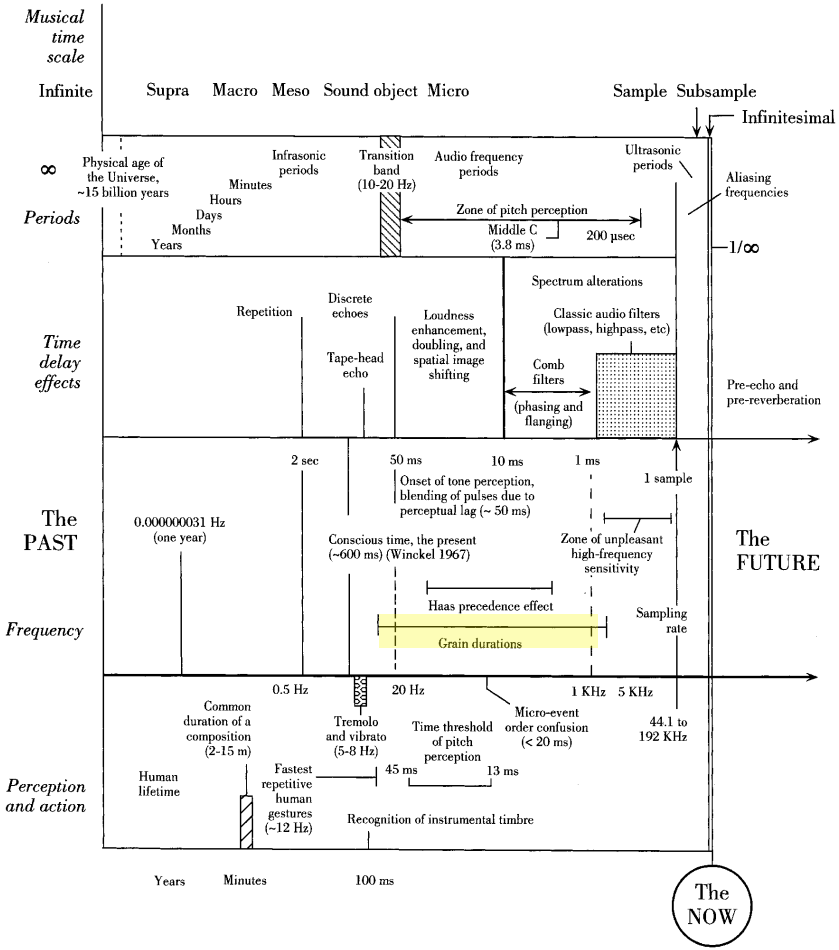
PART II: METHOD

Problem Formalisation
Versio Platform

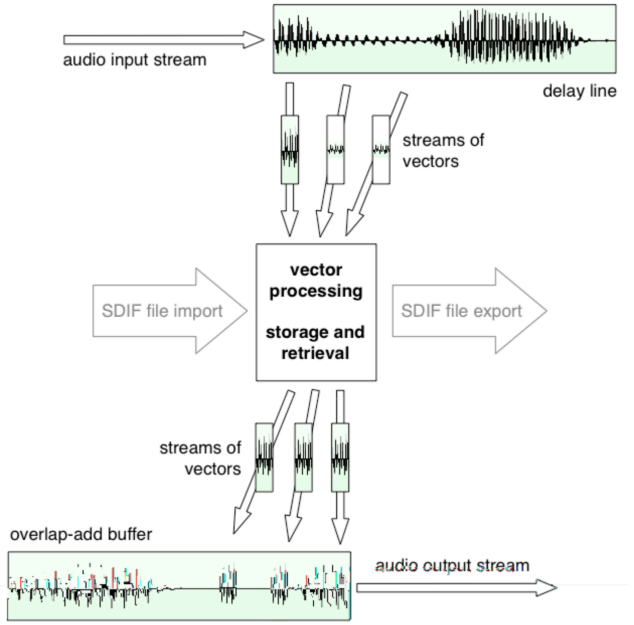
PART III: DEMO

UX Panel Layout / Guide
Demo / Audio Output

MEMORY + GRANULARITY



The time domain, segmented into periods, time delay effects, frequencies, and perception and action (Roads, 2004).



Data flow for a generalised auditory Gabor process (Schnell and Schwarz, 2005).

GRANULAR SYNTHESIS

- Tape-splice / musique concrète approach
- Splices=segments + Genes=grains
- Varispeed playback of 1 segment at a time
- Tape/splice source, triggered capture



Make Noise Morphagene (2019)

- Roads-style granular synthesis / grain-clouds
- 6 layers feed into 1 grain stream
- Params: intensity, size, spray, pitch, scan



Instruo Arbhar (2021)

- Live granular processor on rolling buffer
- Freeze holds buffer
- Params: position, size, pitch, density, texture



Mutable Instruments Clouds (2014)

- Granular synth/delay effect
- Always-recording rolling buffer
- Freeze + density-driven grain streams



Mutable Instruments Beads (2021)

- 2 sides: micro-looper + spatial wet channel
- Clock knob for tone, length, and quality
- Granular stretch controls length and scan speed.
- Freeze holds buffer
- **MIDI but not modular**
- **1 channel**
- **Destructive layering**



Chase Bliss Mood MK2 (2023).

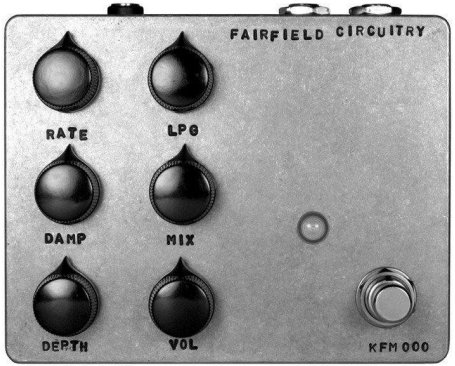
TIMBRAL MODIFICATION



Tascam Portastudio 414 MK2 (1997).



E-mu Emulator II (1984).



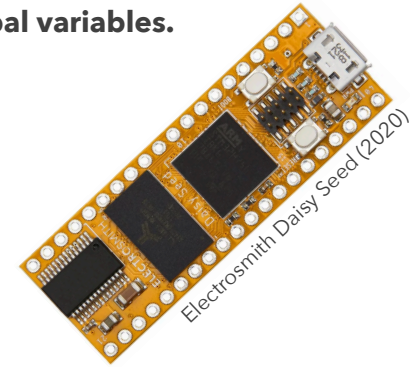
Fairfield Circuitry Shallow Water (2017).

E-mu Systems,
applied magic for the arts
Emulator 2 Output: Sample/Holds
DRAWN BY: D.P.R. | DOC NO. S-6147-001 PAGE 1 OF 15
REV 1.C | DATE 850607 | 7 - 39

PROBLEM FORMALISATION

3-Slot memory device, with UX for parameterisation of slot and global variables.

- **Three interconnected sub-problems inherently arise:**



Sub-problem 1: **CPU / SDRAM / Flash.**

- 3 slots + tape loop / granular stretch + FX on processor; ~1 ms to compute each block
- Dry and wet buffers are always recording, and captured to slots with decimation when triggered.
- **Solution:** Daisy Seed (STM32H750, Cortex-M7 @480 MHz), 64 MB SDRAM.

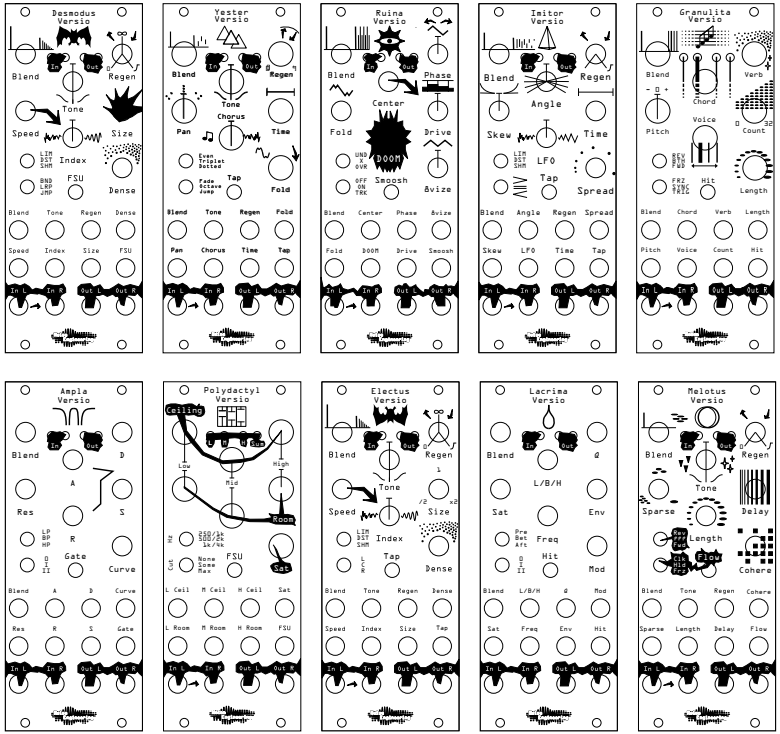
Sub-problem 2: **UX.**

- Eurorack has limited control surface per module; Loads of possible parameters available, choosing the right ones requires testing.
- Visual representation to ensure the user is aware of device state, settings, input and output.
- **Solution:** Button-hold combos + switch modes; capture decoupled from edit allows capture arming of one slot while editing another.

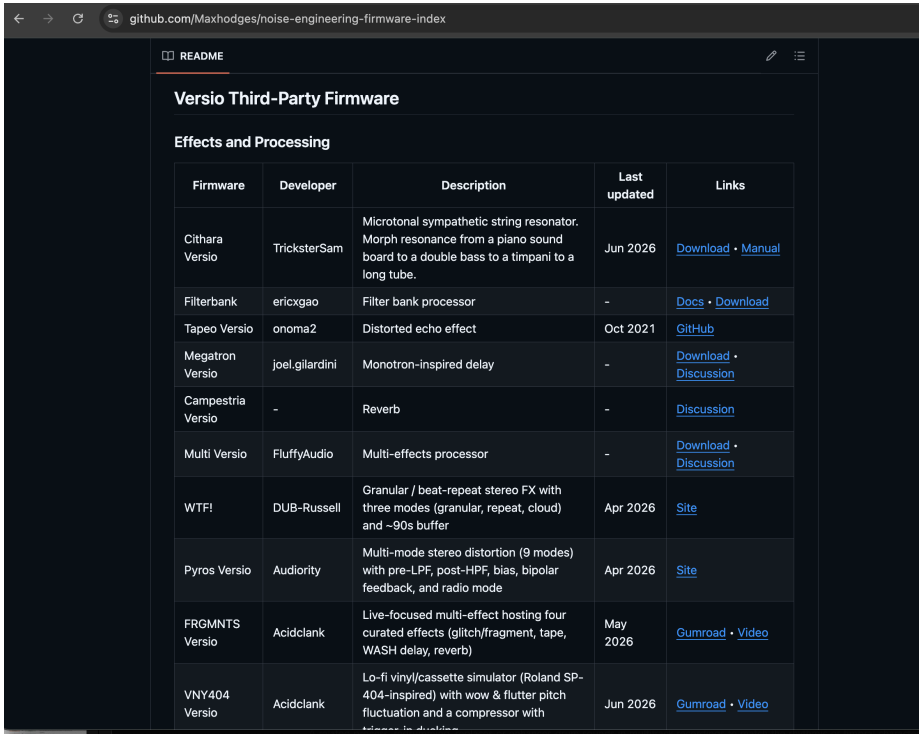
Sub-problem 3: **Algorithmic quality / authenticity.**

- Basic decimation sounds flat and requires character through additional processes (e.g., E-mu μ-law companding).
- **Solution:** Emulate the Emulator.

NOISE ENGINEERING VERSIO PLATFORM



Official Noise Engineering Versio Firmwares (2026).



github.com/Maxhodges/noise-engineering-firmware-index

README

Versio Third-Party Firmware

Effects and Processing

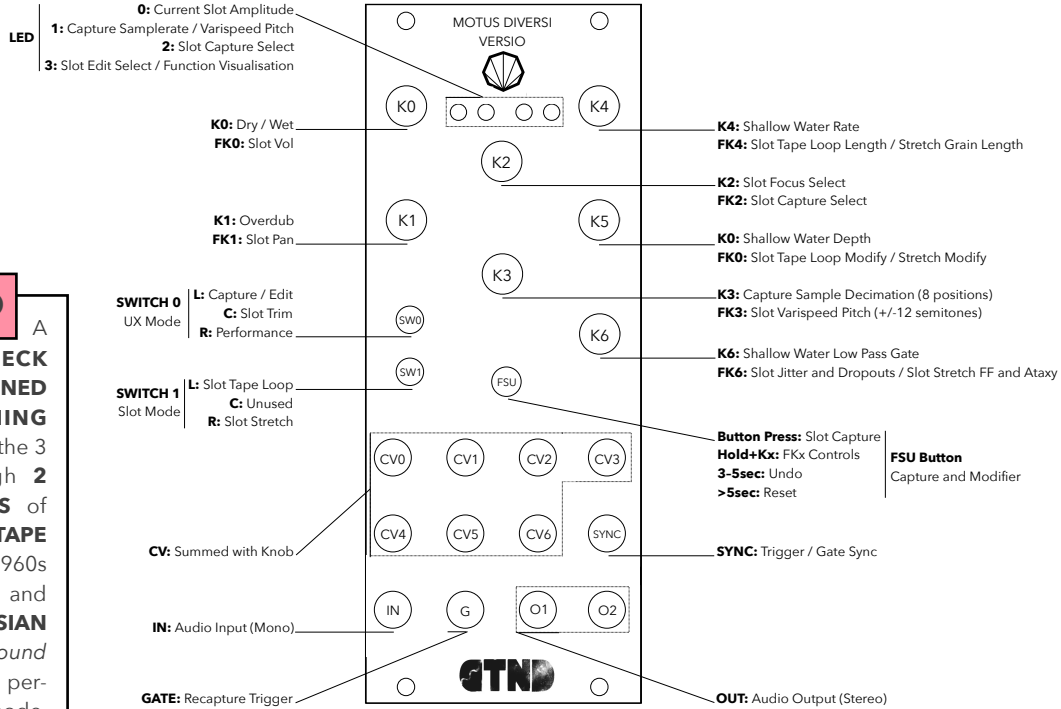
Firmware	Developer	Description	Last updated	Links
Cithara Versio	TricksterSam	Microtonal sympathetic string resonator. Morph resonance from a piano sound board to a double bass to a timpani to a long tube.	Jun 2026	Download • Manual
Filterbank	ericxgao	Filter bank processor	-	Docs • Download
Tapeo Versio	onoma2	Distorted echo effect	Oct 2021	GitHub
Megatron Versio	joel.gilardini	Monotron-inspired delay	-	Download • Discussion
Campestria Versio	-	Reverb	-	Discussion
Multi Versio	FluffyAudio	Multi-effects processor	-	Download • Discussion
WTF!	DUB-Russell	Granular / beat-repeat stereo FX with three modes (granular, repeat, cloud) and ~90s buffer	Apr 2026	Site
Pyros Versio	Audiority	Multi-mode stereo distortion (9 modes) with pre-LPF, post-HPF, bias, bipolar feedback, and radio mode	Apr 2026	Site
FRGMNTS Versio	Acidclank	Live-focused multi-effect hosting four curated effects (glitch/fragment, tape, WASH delay, reverb)	May 2026	Gumroad • Video
VNY404 Versio	Acidclank	Lo-fi vinyl/cassette simulator (Roland SP-404-inspired) with wow & flutter pitch fluctuation and a compressor with	Jun 2026	Gumroad • Video

Third-party Versio firmware repos via Github (accessed July 2026).

PANEL CONTROLS

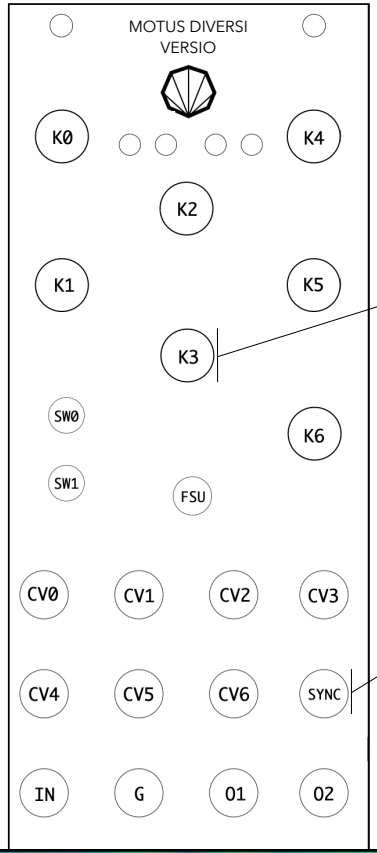
MOTUS DIVERSI VERSIO

A *reeeeeaaallllllyy* compact **3-DECK PERFORMANCE LOOPER DESIGNED FOR CAPTURING AND RUINING SOUND**. Record audio into any of the 3 decks and route each through **2 MODES FROM DISTINCT ERAS** of sonic manipulation. **MODE 1 IS A TAPE LOOPER** designed to emulate 1960s style approach to tape splicing and **MODE 2 IS A MODERN ROADSIAN GRANULAR STRETCH** for *microsound* deconstruction. CV control over per-slot parameters in performance mode. Finally the 3 decks are summed and sent through a Shallow Water engine (pitch warble chorus and low pass gate).



Main panel controls.

SAMPLE RATE DECIMATION / DURATION MODEL

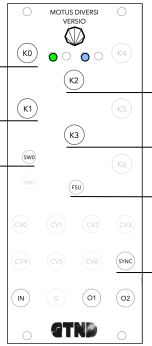


Rung	SRate	Duration: Unsynced (120BPM)	Duration: Synced (170BPM)	Bars
0	48kHz	2s	2.82s	1
1	24kHz	4s	5.65s	2
2	16kHz	6s	8.47s	3
3	12kHz	8s	11.29s	4
4	8kHz	12s	16.94s	6
5	6kHz	16s*	16.94s*	8
6	4kHz	20s*	16.94s*	10
7	3kHz	20s*	16.94s*	12



Mood MK2 clock control (2023).

QUICK START GUIDE



Dry / Wet: 0.5–1.0 (Monitor input)

Overdub=0.0: Dry (no overdub)

SW0=L: Normal/Edit Mode

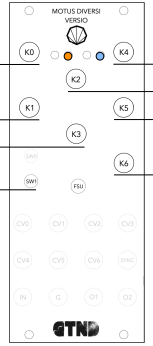
Slot Capture Select (w/FSU-hold)=0.0: Slot 1 (of 3) armed for capture (LED2=blue)

Capture SR Decimation=0.5: Moderate setting (12kHz, 8s duration*)

FSU Button Press: Capture to Slot (LED0=slot trigger activation)

Sync Input: Clock sync capture duration and playback BPM

Step 1: Capture input to slot.



Slot Volume=0.75: +3db from 0.5=unity

Slot Pan=0.5: Centre pan

Varispeed=0.25: Down perfect fifth (LED1=orange).

SW1=R: Granular stretch mode

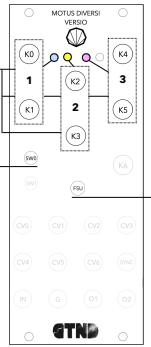
Slot Stretch Length=0.3: Moderate length grain (relative to full slot duration (1.0).

Slot Edit Select (w/o FSU-hold)=0.0: Slot 1 (of 3) set to be edited (LED3=blue).

Slot Stretch Modify=0.55: Just above frozen

Slot Stretch FF/Ataxy=0.55: Full FF (FF=2) and slight spray.

Step 2: Edit per-slot mode and mode parameters. All params accessed with FSU-hold apart from Slot Edit Select.

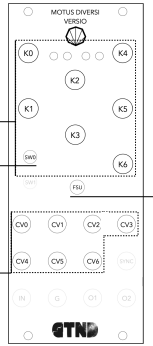


Slot loop start and end pairs
Brightness in LEDs 0-2 indicate relative length of selection

SW0=C (w/FSU-hold):
Audio start and end loop trim mode

FSU Capture disabled. Other functions operational.

Step 3: Edit per-slot mode and parameters.



Potentiometers control FKx without FSU-hold

SW0=L (w/FSU-hold): Performance mode

FSU Capture disabled. Other functions operational.

CV assignments: Adjust per-slot parameters of potentiometers

Step 4: Performance mode exposes FKx per-slot parameters to CV control. FSU-hold not needed.

THANKS FOR LISTENING!

(DEMO)